Lab 07 – Python Tuple

Programming Exercise:

1. Use inbuilt min and max functions to perform the task of getting the minimum and maximum value of in a list of tuples for a particular element position in a tuple.
Sample = [(2, 3), (4, 7), (8, 11), (3, 6)]

Source Code:

```python
sample = [(2, 3), (4, 7), (8, 11), (3, 6)]

# Get minimum and maximum for the first element (index 0)
min_value = min(sample, key=lambda x: x[0])
max_value = max(sample, key=lambda x: x[0])

print("Min value for first element:", min_value)
print("Max value for first element:", max_value)
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\
Min value for first element: (2, 3)
Max value for first element: (8, 11)

Process finished with exit code 0
```

2. A dartboard of radius 10 and the wall it is hanging on are represented using the two dimensional coordinate system, with the board's center at coordinate (0; 0). Variables x and y store the x- and y-coordinate of a dart hit. Write an expression using variables x and y that evaluates to True if the dart hits (is within) the dartboard, and evaluate the expression for these dart coordinates:

(a) (0; 0)

(b) (10; 10)

(c) (6; 6)

(d) (7; 8)

Source Code:

```
def is_within_dartboard(x, y):  1 usage
    # The condition for being within the dartboard
    return x**2 + y**2 <= 10**2

# Test the coordinates
coordinates = [(0, 0), (10, 10), (6, 6), (7, 8)]

for (x, y) in coordinates:
    result = is_within_dartboard(x, y)
    print(f"Coordinate ({x}, {y}) hits the dartboard: {result}")
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\pyt
Coordinate (0, 0) hits the dartboard: True
Coordinate (10, 10) hits the dartboard: False
Coordinate (6, 6) hits the dartboard: True
Coordinate (7, 8) hits the dartboard: False

Process finished with exit code 0
```

3. Write Python expressions corresponding to these statements:

(a) The number of characters in the word "anachronistically" is 1 more than the number of characters in the word "counterintuitive."

- **(a)** `len("anachronistically") == len("counterintuitive") + 1`

(b) The word "misinterpretation" appears earlier in the dictionary than the word "misrep- resentation."

- **(b)** `"misinterpretation" < "misrepresentation"`

(c) The letter "e" does not appear in the word "floccinaucinihilipilification."

- **(c)** `"e" not in "floccinaucinihilipilification"`

(d) The number of characters in the word "counterrevolution" is equal to the sum of the number of characters in words "counter" and "resolution."

- **(d)** `len("counterrevolution") == len("counter") + len("resolution")`

4. Write a program in Python that holds an empty tuple and fill that tuple after taking user input for names of provinces of Pakistan n fill an empty tuple and print.

Source Code:

```python
provinces = ()
num_provinces = int(input("How many provinces of Pakistan would you like to enter? "))
# Loop to take input for each province and add it to the tuple
for _ in range(num_provinces):
    province = input("Enter the name of a province: ")
    provinces += (province,)

# Print the tuple containing the provinces
print("The provinces of Pakistan you entered are:", provinces)
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjects\main\main 3.py"
How many provinces of Pakistan would you like to enter? 4
Enter the name of a province: sindh
Enter the name of a province: punjab
Enter the name of a province: balochistan
Enter the name of a province: KPK
The provinces of Pakistan you entered are: ('sindh', 'punjab', 'balochistan', 'KPK')

Process finished with exit code 0
```

5. Start by assigning to variables monthsL and monthsT a list and a tuple, respectively, both containing strings 'Jan', 'Feb', 'Mar', and 'May', in that order. Then attempt the following with both containers:

(a) Insert string 'Apr' between 'Mar' and 'May'.
(b)Append string 'Jun'.
(c) Pop the container.
(d) Remove the second item in the container.
(e)Reverse the order of items in the container.
(f)Sort the container.

Note: when attempting these on tuple monthsT you should expect errors.

Source Code:

```
# Create a list and a tuple with the same elements
monthsL = ['Jan', 'Feb', 'Mar', 'May']
monthsT = ('Jan', 'Feb', 'Mar', 'May')

#(a)
    # FOR LIST
monthsL.insert( __index: 3,  __object: 'Apr')
print("After inserting 'Apr':", monthsL)
   # FOR TUPLE
# This will raise an error
# monthsT.insert(3, 'Apr')

#(b)
   # FOR LIST
monthsL.append('Jun')
print("After appending 'Jun':", monthsL)
   # FOR TUPLE
# This will raise an error
# monthsT.append('Jun')

#(c)
   # FOR LIST
last_item = monthsL.pop()
print("After popping:", monthsL)
print("Popped item:", last_item)
   # FOR TUPLE
# This will raise an error
# monthsT.pop()

```

```python
#(d)
   # FOR LIST
del monthsL[1]  # Alternatively, monthsL.pop(1)
print("After removing the second item:", monthsL)
   # FOR TUPLE
# This will raise an error
# del monthsT[1]

#(e)
   # FOR LIST
monthsL.reverse()
print("After reversing:", monthsL)
   # FOR TUPLE
# This will raise an error
# monthsT.reverse()

#(f)
   # FOR LIST
monthsL.sort()
print("After sorting:", monthsL)
   # FOR TUPLE
# This will raise an error
# monthsT.sort()
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\
After inserting 'Apr': ['Jan', 'Feb', 'Mar', 'Apr', 'May']
After appending 'Jun': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
After popping: ['Jan', 'Feb', 'Mar', 'Apr', 'May']
Popped item: Jun
After removing the second item: ['Jan', 'Mar', 'Apr', 'May']
After reversing: ['May', 'Apr', 'Mar', 'Jan']
After sorting: ['Apr', 'Jan', 'Mar', 'May']

Process finished with exit code 0
```